DIP BHATTACHARJEE

Computer Science & Engineering Undergraduate | AI/ML Research Enthusiast

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dipjee678.github.io/2nd-Protfolio

PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate majoring in **Computational Theory**, with a strong interest in Artificial Intelligence, Machine Learning, Natural Language Processing, Deep Learning, and applied research. Academic training includes Machine Learning, Natural Language Processing, Graph Theory, Data Structures and Algorithms, and Object-Oriented Programming. Experienced in programming, software development, AI/ML-based projects, and solving computational problems. Passionate about developing practical, data-driven solutions and continuously expanding technical expertise in AI/ML, NLP, and emerging intelligent technologies.

EDUCATION

Bachelor of Science in Computer Science and Engineering — **Aug 2023 – Present**
American International University-Bangladesh — *CGPA: 3.83/4.00*

Major: Computational Theory

Completed Major Courses: Natural Language Processing, Machine Learning, Basic Graph Theory.

Additional Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming with Java, Programming in C++, and Discrete Mathematics.

Higher Secondary Certificate (HSC) — **2020 – 2022**
Mohipal Government College – Science — *GPA: 4.89/5.00*

Secondary School Certificate (SSC) — **2018 – 2020**
Batisa Secondary School – Science — *GPA: 4.89/5.00*

RESEARCH & CONFERENCE PAPERS

Progressive Two-Stage Transfer Learning with Dual Imbalance Compensation for Interpretable Enamel Caries Classification
Accepted for Presentation at the 5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)

Development of a Cloud-Linked Low-Cost Machine Learning-Enabled Weather Monitoring System
Published in the Proceedings of IEEE TENSYMP 2026 · DOI: 10.1109/TENSYMP68909.2026.11704037

Diagnostic-Guided Monte Carlo Risk Estimation Using GARCH, EGARCH, and Bootstrap Models
Accepted for Presentation at the 3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)

TECHNICAL SKILLS

Programming:	Python, C, C++, Java, C#, JavaScript
AI & Machine Learning:	Machine Learning, Deep Learning, Transfer Learning, Classification, Model Training & Evaluation
Natural Language Processing:	Text Preprocessing, Text Classification, Feature Extraction, NLP Fundamentals
Data & ML:	Data Preprocessing, Data Analysis, Model Evaluation, Class Imbalance Handling
AI Applications:	LLM APIs (Google Gemini), Structured Outputs, Prompt Design, Multilingual Text Generation, Serverless Deployment (Vercel)
Core CS:	Data Structures & Algorithms, Graph Theory, Object-Oriented Programming, Problem Solving
Research Areas:	Computer Vision, Medical Image Analysis, Explainable AI, NLP, IoT-based Intelligent Systems
Tools:	Microsoft Office, AutoCAD, Adobe Photoshop

PROJECTS

LeadDesk: AI Enquiry CRM for Bangla, Banglish and English *JavaScript, Gemini API, Vercel*

- Built a web CRM that turns customer messages from WhatsApp, Facebook and web forms, written in Bangla, Banglish or English, into structured leads using an LLM with schema-enforced (Zod) structured outputs.
- Classifies each message (sales, support, spam, job seeker) and drafts replies in the customer's own language, grounded only in the company's facts.
- Added a drag-and-drop sales pipeline, 48-hour follow-up reminders, duplicate merging, an AI weekly summary, and a keyword fallback that works without an API key; deployed with Vercel serverless functions.
- Live demo: biai-crm-eight.vercel.app · Source: github.com/dipjee678/LeadDesk

Hospital Management System *Java*

- Developed a system to support essential hospital operations, including patient records, appointments, billing, and staff management.
- Applied object-oriented programming principles in Java to design structured and maintainable system functionality.

Poultry Farm Management System *Java, Database*

- Developed a database-driven management system to organize and manage poultry farm-related information efficiently.
- Applied Java programming and database concepts to implement structured data storage, management, and system operations.

EXPERIENCE

Instructor *NextGen Coaching Center for Admission Preparation*

- Taught key academic subjects required for university admission preparation, helping students strengthen fundamental concepts.
- Developed mock tests and quizzes to improve students' problem-solving speed, accuracy, and examination preparation.
- Conducted one-on-one mentoring sessions to help students understand challenging topics and improve their academic confidence.

Freelance Tutor *Programming & Problem Solving*

- Mentored students in Python, JavaScript, and C++, strengthening programming fundamentals and problem-solving skills.
- Designed personalized learning plans covering programming, data structures, algorithms, and computational problem solving.
- Guided students through practical coding exercises and project-based learning activities.

ACHIEVEMENTS & CERTIFICATIONS

- **Dean's Award**

ADDITIONAL INFORMATION

Languages: Bangla (Native), English (Fluent), Hindi (Proficient in Listening, Basic Speaking)
Research Interests: Artificial Intelligence, Machine Learning, Natural Language Processing, Deep Learning, Computer Vision, Explainable AI, Medical Image Analysis, and Intelligent Systems